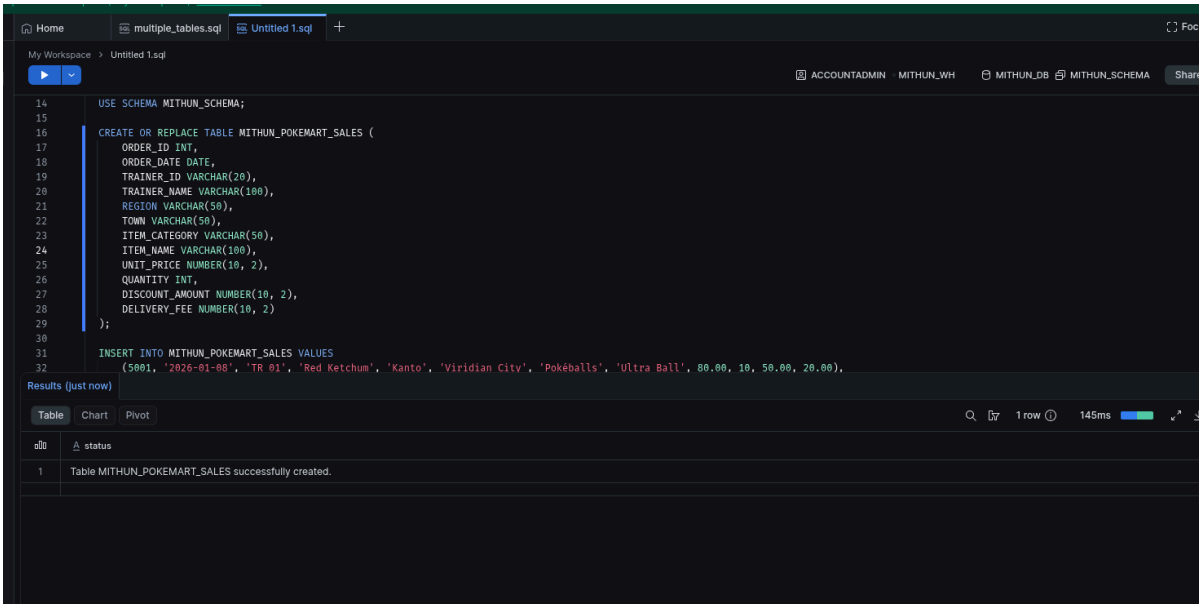
Snowflake to Power BI Integration & Dashboard Creation

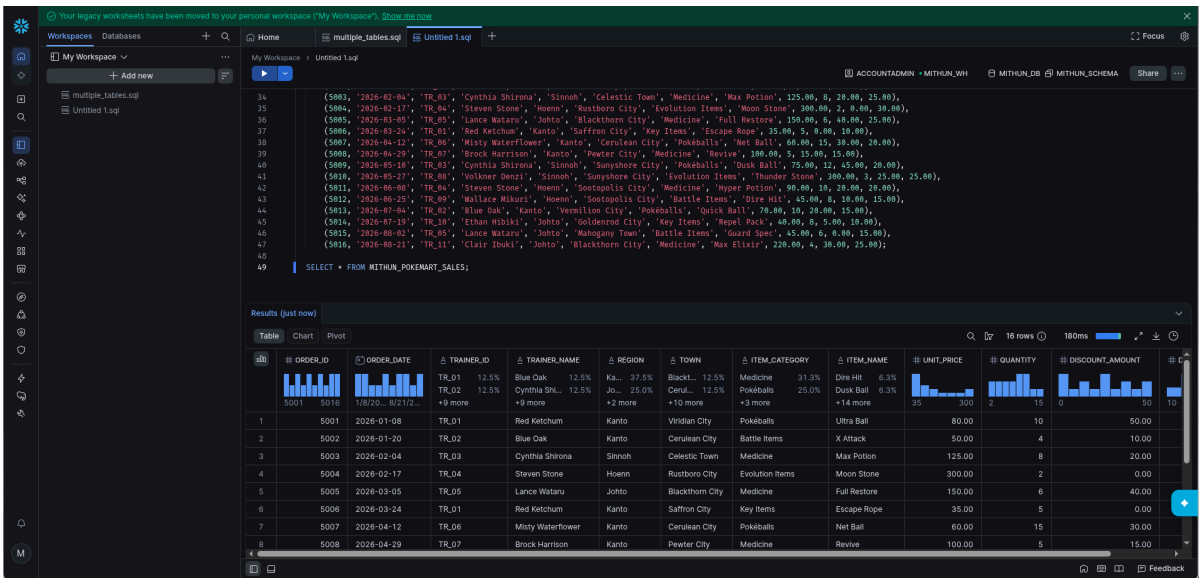
Step 1: Snowflake Environment Setup & Data Ingestion

Objective: Initialize a dedicated virtual warehouse, database, and schema in Snowflake, and populate a raw table with sample transaction data.

Description:

1. Open a new SQL Worksheet in the Snowflake UI.
2. Execute the setup script to provision the infrastructure and load the data.
3. Verify the data using a SELECT query.
4. Note down the warehouse name (MITHUN_WH) and Server URL.



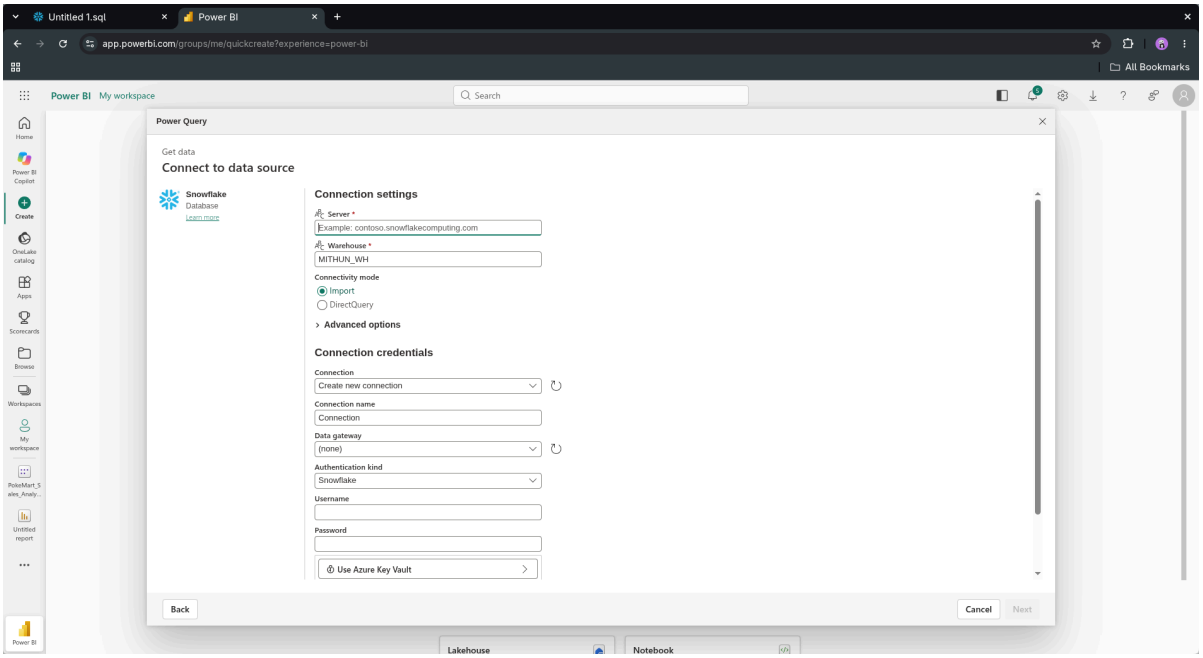


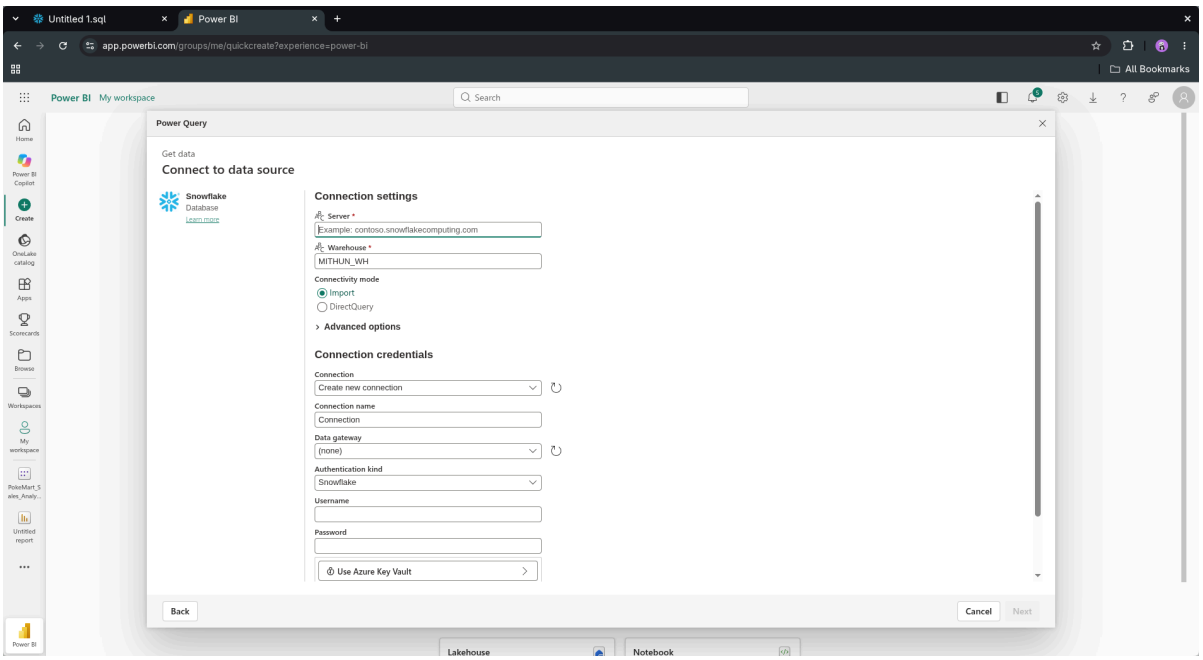
Step 2: Establish Power BI Connection

Objective: Securely connect Power BI Desktop to the Snowflake data warehouse.

Description:

1. Open Power BI Desktop and select Get Data > Snowflake.
2. Input the Server URL and Warehouse name (MITHUN_WH).
3. Authenticate and select the MITHUN_POKEMART_SALES table.



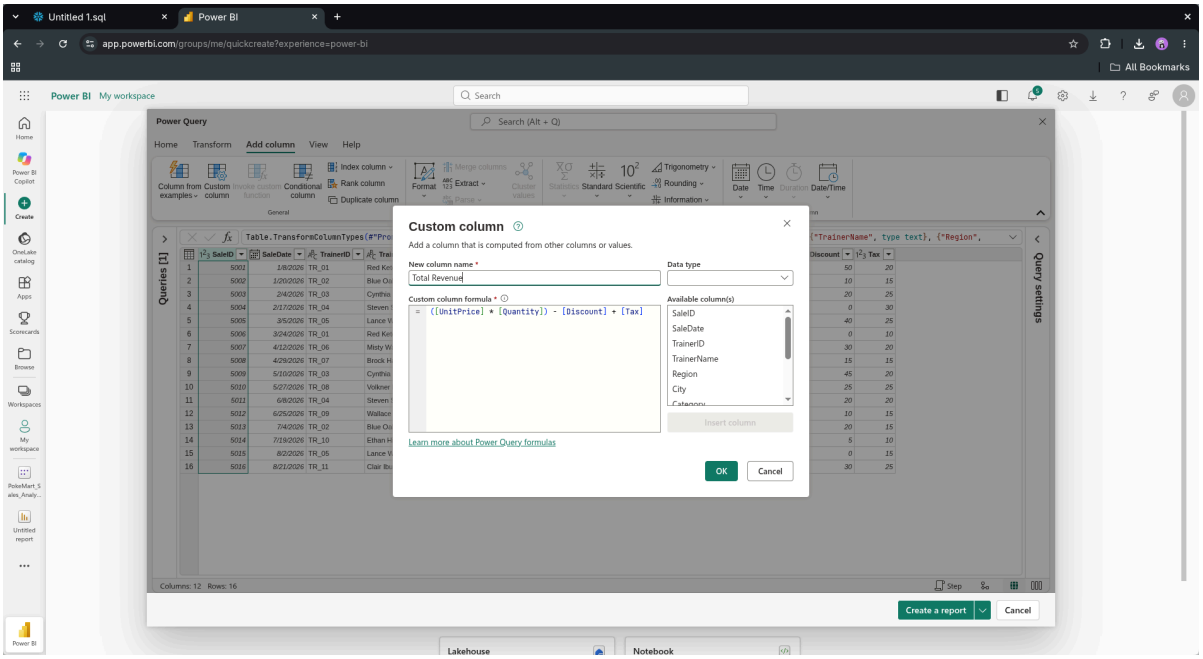


Step 3: Data Transformation in Power Query

Objective: Enrich the dataset by calculating aggregated financial metrics.

Description:

1. In Power Query Editor, go to Add Column > Custom Column.
2. Name it Total Revenue with formula: = ([UnitPrice] * [Quantity]) - [Discount] + [Tax].
3. Ensure the data type is a numeric value.
4. Click Close & Apply.

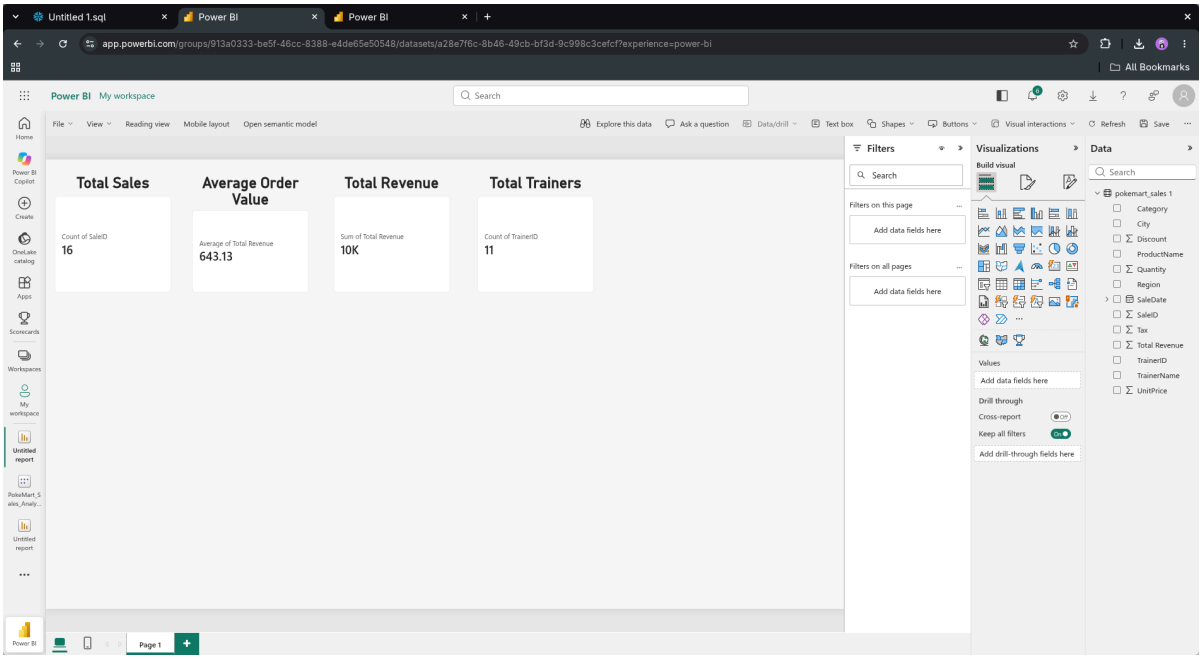


Step 4: High-Level KPI Cards Configuration

Objective: Provide immediate visibility into core business metrics using Card visuals.

Description:

1. Create 4 Card visuals for Total Sales (Count Distinct of SaleID), Total Revenue (Sum), Total Trainers (Count Distinct), and Average Order Value (Average).
2. Format titles and adjust font sizes to be uniform (e.g., DIN size 29).



Step 5: Core Analytical Visuals & Slicer Setup

Objective: Build interactive visualizations to analyze revenue trends across categories, geographical regions, and timeframes.

Description:

1. Create a Stacked Bar Chart for Revenue by Category.
2. Create a Line Chart for Revenue Trend by Month.
3. Create a Donut Chart for Revenue Share by Region.
4. Add a Slicer for the Region field.

Step 6: Supplementary Data Visualizations

Objective: Expand analytical depth by introducing 5 additional granular visuals.

Description:

1. Treemap: Revenue Concentration by City.
2. Clustered Column Chart: Items Sold by Category.
3. Table: Top Performing Items sorted by Total Revenue.

4. Area Chart: Total Discounts over Time.
5. Card: Total Items Sold (Sum of Quantity).

